C — Deliverable 3 Document

Project Title: Library Management System
Course: CS3009 — Software Engineering
Deliverable: Assignment 3 — Final Report
Team Name: Stack Architects

Team Members:

- Ali Bin Salman
- Muhammad Rohaan
- Muhammad Atyab

Submission Date: 24-4-26

1. Introduction

The Library Management System (LMS) is a full-stack web application designed to automate and streamline library operations for students, librarians, and administrators.

- **Sprint 1:** Established account management, including student registration, admin approval, role-based login, and account deletion.
- **Sprint 2:** Added book catalog, search, borrowing workflow, approval/rejection, due date assignment, borrow history, and borrow limit enforcement.
- **Sprint 3:** Completed the remaining advanced modules: book returns, automatic fine calculation, fine management, renewal/reissue, reservation/waitlist system, and admin reports. Sprint 3 also included **final integration, UI improvements, and finishing touches** to ensure a complete and fully functional system.

By the end of Sprint 3, the LMS became a **fully integrated library management solution**, covering the entire lifecycle from registration to borrowing, returning, fines, renewal, reservations, and reporting.

2. Project Vision

The vision of the LMS is to provide a **secure, efficient, and user-friendly platform** for library management. Sprint 3 enhanced the system to:

- Automate return and overdue fine processing.
- Enable students to renew and reserve books.
- Provide librarians with comprehensive fine and reservation management.

- Give administrators accurate reports for library activity.
- Ensure the system is fully integrated, consistent, and ready for deployment.

3. Intended Use of the System

Students:

- Search books, request loans, track active loans.
- Renew loans and reserve unavailable books.
- View fines and payment status.

Librarians:

- Manage books, approve/reject borrow requests, record returns.
- Track overdue books, manage fines, and handle reservation queues.

Administrators:

- Approve or reject accounts, manage user roles.
- View reports on issued books, overdue loans, fines, and system activity.

Stakeholder Updates:

- No new stakeholders were added in Sprint 3.
- Student, Librarian, and Admin workflows were enhanced to support the new modules.

4. Features and Functionality

Sprint	**Features Completed**
Sprint 1	Registration, admin approval/rejection, role-based login, account deletion request
Sprint 2	Book catalog management, search, borrow request workflow, approval/rejection, due date assignment, borrow history, borrow limits
Sprint 3	Book Return Module, Automatic Fine Calculation, Fine Management, Renewal/Reissue, Reservation/Waitlist, Admin Reports, Final Integration and UI polishing

Sprint 3 Enhancements:

- Completed the borrowing lifecycle with returns, fines, and renewal modules.
- Ensured fair allocation with reservation queue management.
- Provided administrators with real-time reporting.
- Integrated all modules for a smooth end-to-end workflow.

5. User Stories (Sprint 3)

- As a Librarian, I want to record book returns so that issued books become available.
- As a System, I want to calculate overdue fines automatically so that penalties are applied accurately.
- As a Student, I want to view my fines so that I can check my outstanding balance.
- As a Librarian, I want to mark fines as paid or waived to maintain accurate records.
- As a Student, I want to renew borrowed books so that I can extend my loan period.
- As a Student, I want to reserve unavailable books to receive them when returned.
- As a Librarian, I want to manage reservation queues so that books are assigned fairly.
- As an Admin, I want to view overdue, issued, fine, and activity reports to monitor library operations.

6. Non-Functional Requirements (NFRs)

Performance:

- The system shall respond to page requests within 2–3 seconds.
- Return, renewal, and reservation operations shall execute without delays.

Usability:

- User dashboards shall be intuitive and clearly display available actions.
- Success and error messages shall appear after relevant actions.

Reliability:

- Duplicate returns, fines, or reservations shall be prevented.
- Inventory and user account data shall remain consistent.

Security:

- Only authenticated users may access protected pages.
- Access to Admin, Librarian, and Student pages shall be restricted by role.
- CSRF protection is enabled for all POST actions.

Scalability:

- Database design supports multiple users, books, borrow records, fines, and reservations.
- The system is designed to accommodate future module expansions and reporting features.

7. Testing and Quality Assurance

Test Plan:

- Focused on Return, Fine, Renewal, Reservation, and Admin Reports modules.
- Functional, regression, UI, and integration testing were conducted.
- Test environment: Spring Boot 3.5, Java 17, MySQL, Thymeleaf, localhost:8081.

Black-Box Test Cases (10):

- Loan return, duplicate return, overdue fine calculation, student fine view, fine management, renewal allowed and blocked scenarios, reservation creation, admin report accuracy.

White-Box Test Cases (10):

- Branches and conditions for return, fine calculation, fine management, renewal rules, reservation creation and READY/EXPIRED logic.

Bug Report Summary:

- All critical Sprint 3 defects were identified and fixed.
- Key fixes included duplicate return validation, fine calculation corrections, renewal rule enforcement, and reservation queue management.

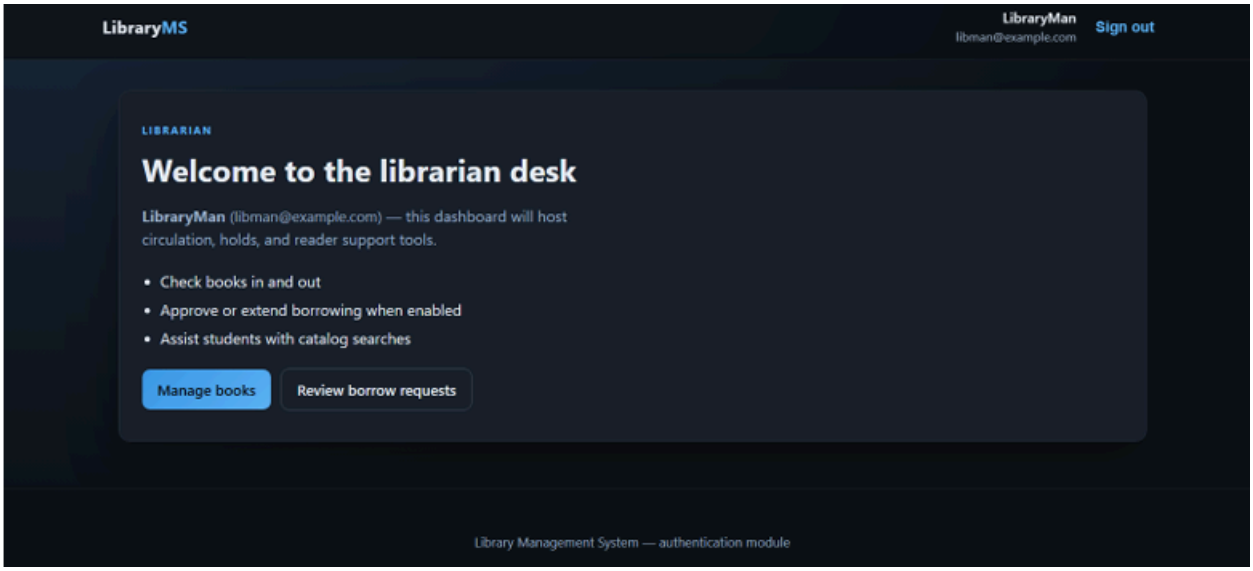
Test Results and Observations:

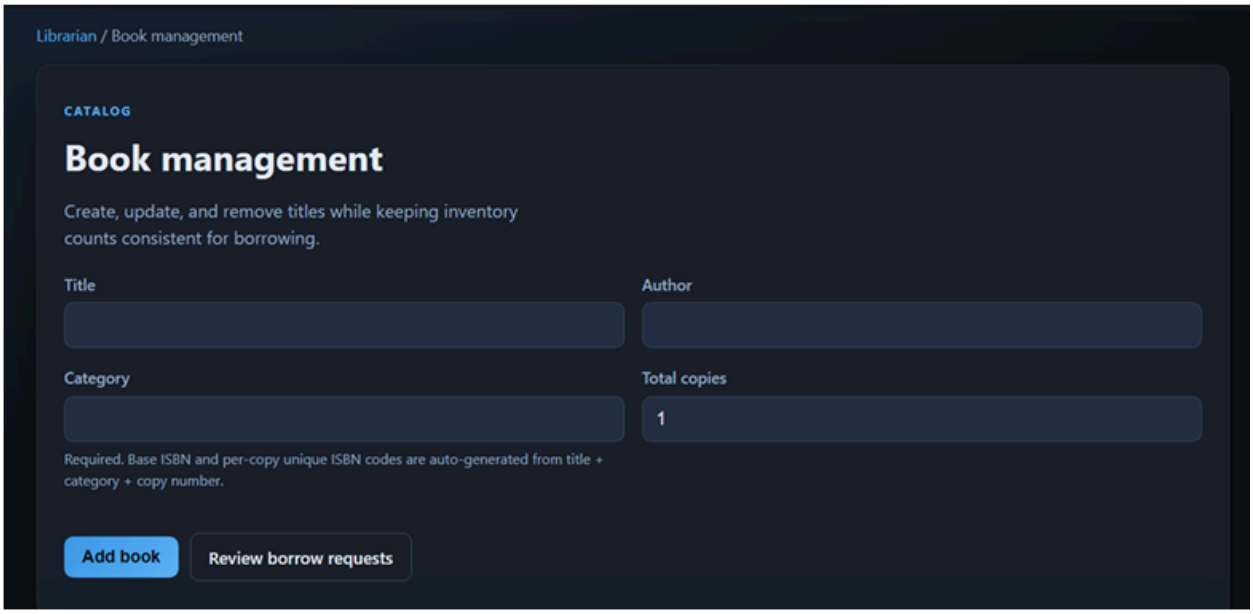
- 100% of selected black-box and white-box tests passed.
- Regression tests confirmed no disruption of Sprint 1 or 2 functionality.
- System correctly handles all end-to-end workflows for Sprint 3 modules.

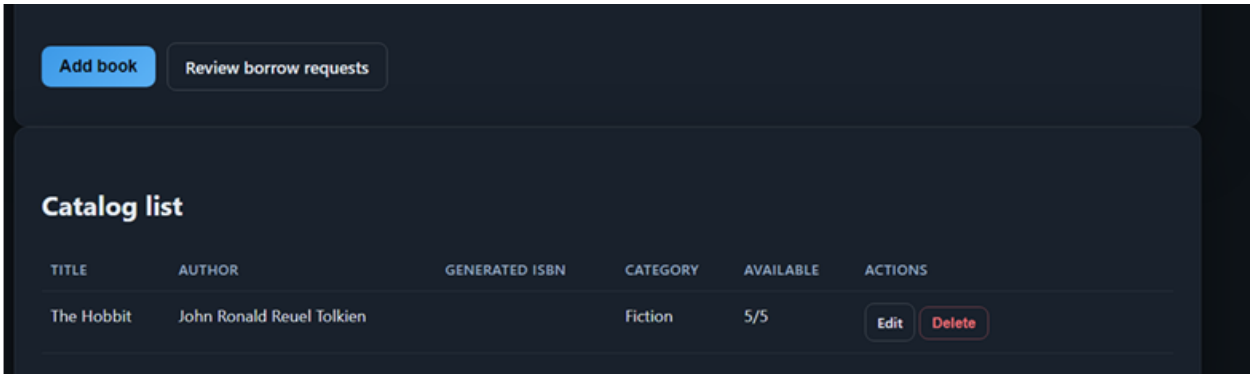
How Testing Improved Quality:

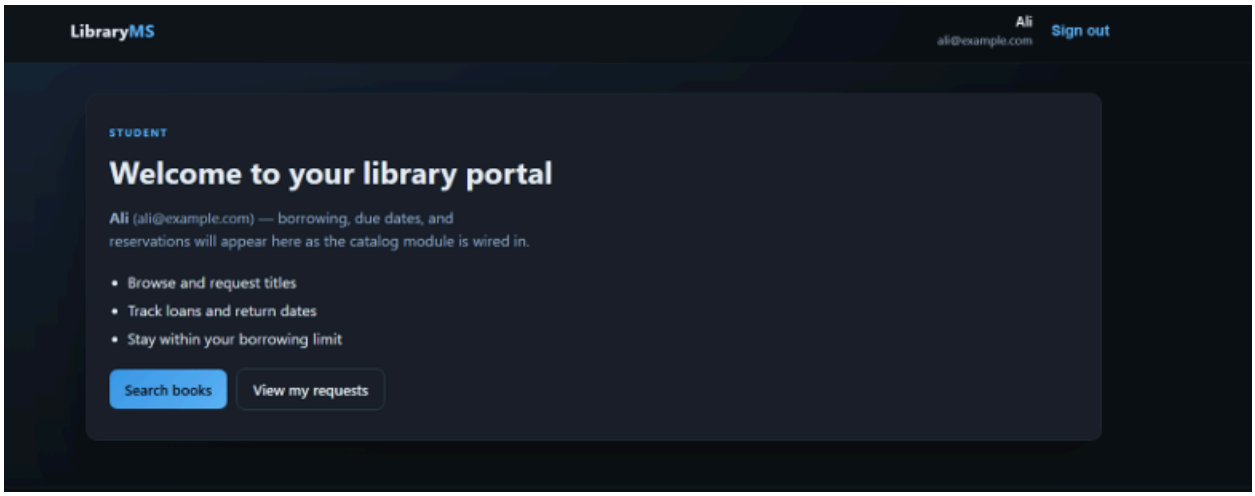
- Ensured module logic correctness, data consistency, and adherence to business rules.
- Validated system integration across all three sprints.

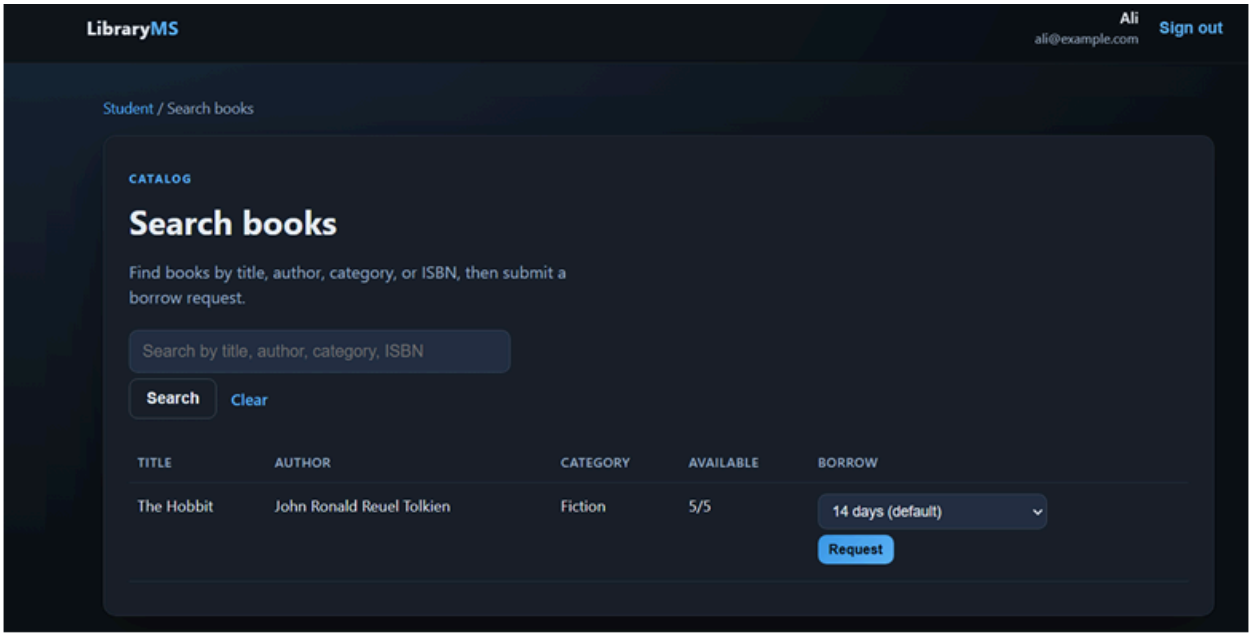
8. Iteration 3 Implementation Screenshots

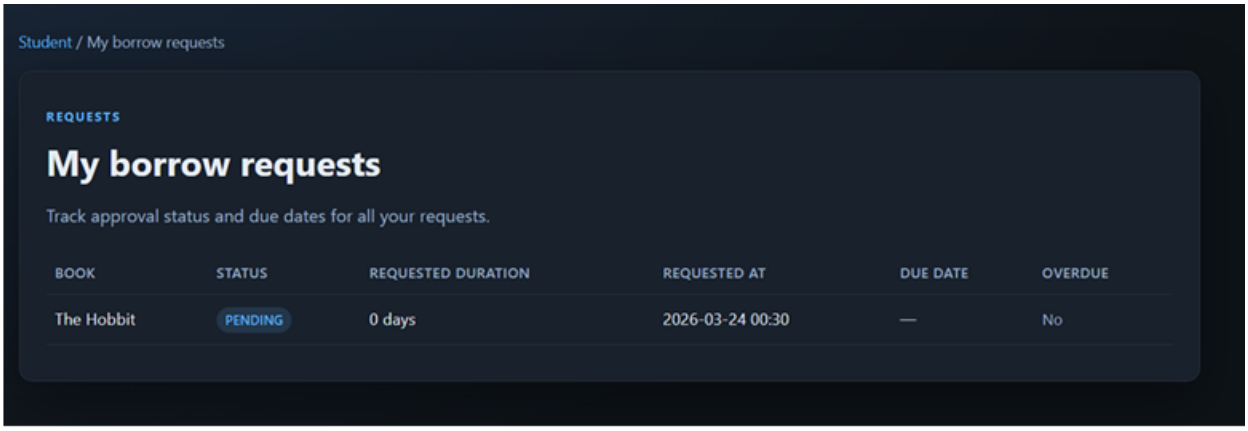


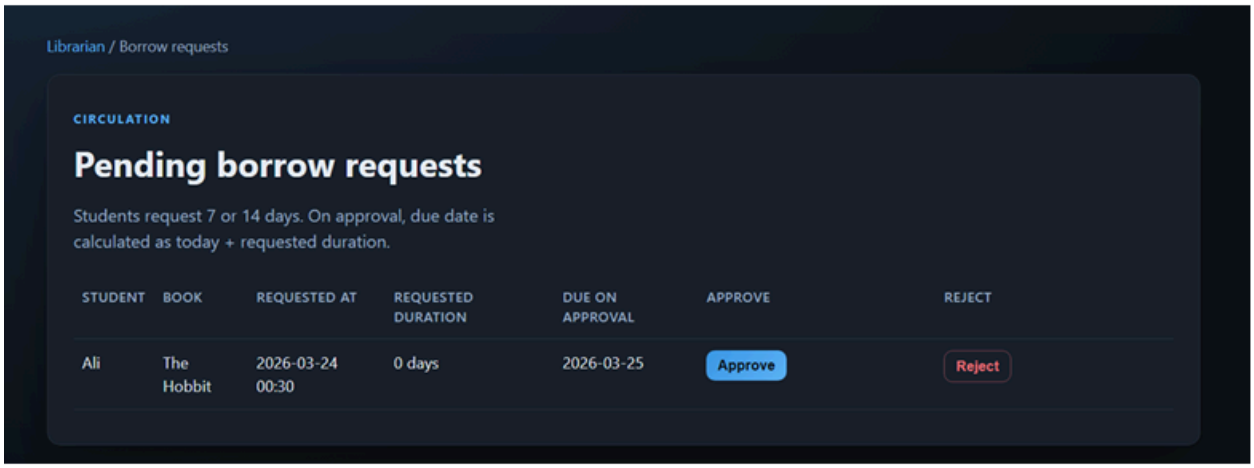



Add book
Review borrow requests
Catalog list
TITLE
AUTHOR
GENERATED ISBN
CATEGORY
AVAILABLE
ACTIONS
The Hobbit
John Ronald Reuel Tolkien
Fiction
5/5
Edit
Delete


LibraryMS
Ali
ali@example.com
Sign out
STUDENT
Welcome to your library portal
Ali (ali@example.com) — borrowing, due dates, and reservations will appear here as the catalog module is wired in.
Browse and request titles
Track loans and return dates
Stay within your borrowing limit
Search books
View my requests


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Ali
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Sign out
Student / Search books
CATALOG
Search books
Find books by title, author, category, or ISBN, then submit a borrow request.
Search by title, author, category, ISBN
Search
Clear
TITLE
AUTHOR
CATEGORY
AVAILABLE
BORROW
The Hobbit
John Ronald Reuel Tolkien
Fiction
5/5
14 days (default)
Request





9. Work Division for Sprint 3

Ali Bin Salman:

- Led final integration and Sprint 3 planning
- Developed return and fine management requirements
- Verified dashboards and route access
- Coordinated documentation and report preparation

Muhammad Rohaan:

- Implemented backend logic for return, fine, renewal, and reservation modules
- Ensured transactional safety and service-layer logic
- Developed admin reports queries
- Maintained Scrum board and tracked task progress

Muhammad Atyab:

- Designed and implemented Sprint 3 UI pages
- Performed black-box and white-box testing
- Captured screenshots for the report
- Verified bug fixes and regression testing

Collaborative Work:

- Regression testing across all three sprints
- Integration of Sprint 3 modules with previous features
- Final UI polishing and defect fixing
- Documentation preparation for deliverable submission

10. Final System Summary

Across all three sprints, the LMS evolved into a **complete, integrated system**:

- **Sprint 1:** Account management and authentication
- **Sprint 2:** Book catalog, search, borrowing workflow, and history
- **Sprint 3:** Return, fine, renewal, reservation, admin reporting, and final integration

The system now provides end-to-end support for library operations: registration, borrowing, return, fines, renewal, reservations, and administrative monitoring. It is stable, fully functional, and aligned with the original project scope.